

# VEDA AI ASSESSMENT - MID-TERM PHYSICS

Class 11 | Time: 2 Hours | Max Marks: 40

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## Q 1. [5 Marks]

State Newton's Second Law of Motion and derive the vector equation  $F = ma$ .

## Q 2. [5 Marks]

Calculate the kinetic energy of a 4 kg object moving at a uniform velocity of 10 m/s.

## Q 11 (a). [5 Marks]

State Ohm's Law and explain the factor of resistance in a conductive wire.

## Q 11 (b). [5 Marks]

Derive the formula for equivalent resistance  $1/R_{eq} = 1/R_1 + 1/R_2$  in parallel.

## Q 3. [5 Marks]

Explain the Photoelectric Effect and write Einstein's photoelectric equation.

## Q 4. [5 Marks]

Define Total Internal Reflection and state the critical angle condition with formula.

## Q 5. [10 Marks]

Describe the working principle of a transformer, write the turns ratio  $V_s/V_p = N_s/N_p$ , and detail major energy losses (copper loss, eddy currents, hysteresis).